Redefining Health

Designing a Resistive, Omnidirectional Surface for a Gamified Fitness Experience

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Traditional Treadmill

Traditional treadmills allow a user to move indefinitely. However, as they only support unidirectional movement, their effectiveness in various industries is limited.

Omnidirectional treadmill

Omnidirectional treadmills enable movement in any direction. They have increased physiological utility, especially when combined with virtual reality technology to create immersive experiences.

APPLICATIONS AND USE CASES

LIMITATIONS OF CURRENT MARKET

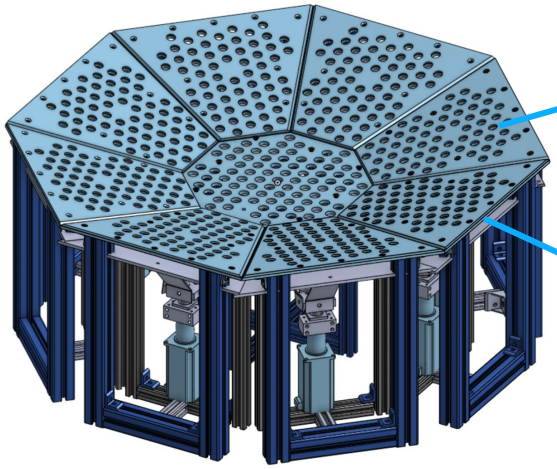
- Expensive due to components like motors & sensors
- Large in size and take up a lot of space
- Require non-intuitive walking that causes motion sickness

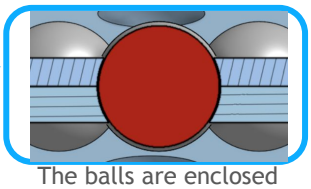
OUR PROJECT REQUIREMENTS

- Must allow movement without motor actuation
- Prioritize modularity, ease of use, and user safety
- Should improve on market limitations

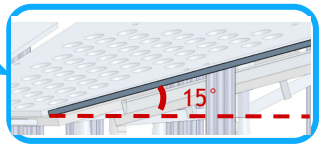
OUR TREADMILL

Comprised of a center base plate and eight angled plates enclosing balls that can freely rotate. The user can step in any direction and will slide to a neutral position in the center.

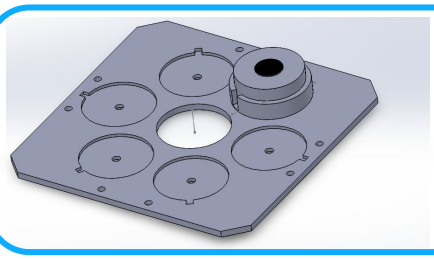




The balls are enclosed between two plates with a spherical cutout.



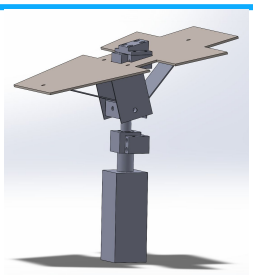
Side plates are angled & use gravity to slide the user to center



MAGNET BRAKING SYSTEM - CENTER PLATE

The **center plate** is controlled by a bed of electromagnets that are able to stop ball motion at rates depending on the current applied. This system allows for more precise control, a shorter center profile, and greater support.

FRICTION BRAKING SYSTEM - SIDE PLATES

The **angled plates** are controlled by a rubber material that applies a friction force to the balls when the platform it is fixed to is risen and lowered by a motor.

While less precise, this system is cost-effective and efficient as the user spends majority of their use on the center plate.